CSE320: Data Communications

Assignment 2 [Summer 2025]

Full Marks: 20

YOU MUST SUBMIT HARDCOPY OF THE ASSIGNMENT

For this assignment, consider **X** to be the 6-bit binary representation of the last two digits of your ID % 64. For example, if your ID is 22101075, then **X = 75 % 64 = 11 = 001011**. If, however, you have any special ID for which **X** becomes something like 111111, 000000, or any other stream where there is only ONE 0 or 1 (e.g. 000001, 010000, 111101, 111110, etc), then use **X = 100110**.

Analog Transmission

Amplitude Shift Keying (ASK)

1. Determine the digital bit stream from the analog signal below. The signal was modulated using Binary ASK, where 0 means signal element with no amplitude and 1 means signal element with amplitude of 3V.

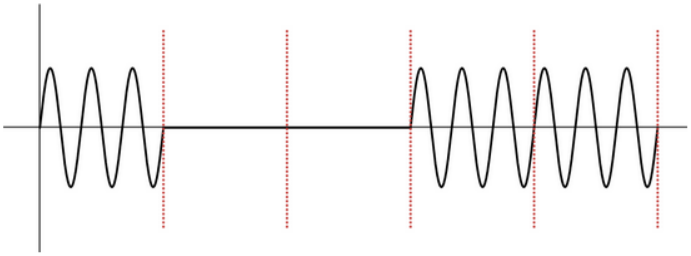


Figure 1: Modulated signal for Question 1

2. Draw the analog signal for the digital bit stream **X** using Binary ASK, where 0 means signal element with amplitude of 3V and 1 means signal element with amplitude of 6V. [For each signal element, frequency = 2, phase = 0°]

3. In a Multi-level ASK, for each signal element, we want to send 2 bits at a time. We have used a carrier signal that has a frequency of 10 Hz (Each signal element has 2 cycles) and phase is 0°. Amplitude for each 2-bits pattern is given as: 00 → 5V, 01 → 10V, 10 → 15V, 11 → 20V. Based on this, draw the modulated signal for the bit sequence **1001010111**.

4. For the following Multi-level ASK, find the bitstream from the signal below:

Bit Pattern	Amplitude
00	1V
01	3V
10	2V
11	4V

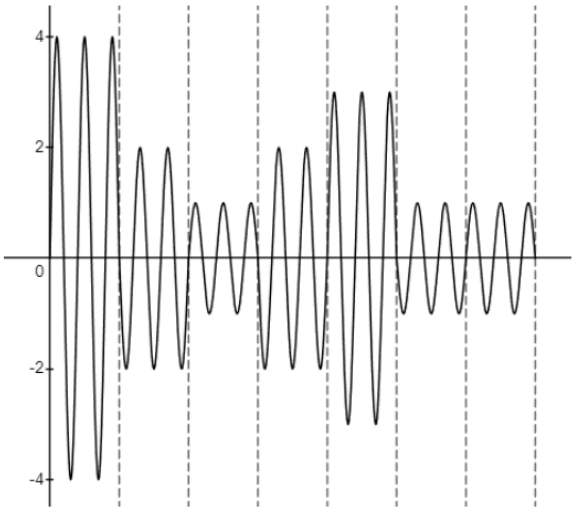


Figure 2: Modulated signal for Question 4

Frequency Shift Keying (FSK)

5. Determine the digital bit stream from the analog signal below. The signal was modulated using Binary FSK, where 0 means signal element with frequency of 2 and 1 means signal element with frequency of 4.

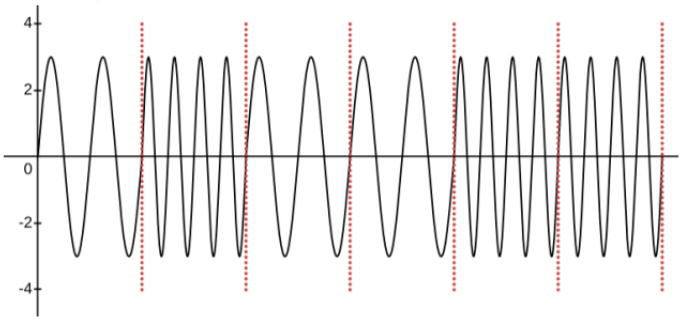


Figure 3: Modulated signal for Question 5

6. Draw the analog signal for the digital bit stream **X** using Binary FSK, where 0 means signal element with frequency of 2 and 1 means signal element with frequency of 4. [Amplitude = 3V, phase = 0°]

7. In a Multi-level FSK, for each signal element, we want to send 2 bits at a time. We have used a carrier signal that has an amplitude of 10V and phase is 0°. Frequency for each 2-bits pattern is given as: 00 → 2, 01 → 4, 10 → 1, 11 → 3. Based on this, draw the modulated signal for the bit sequence **1001010111**.

8. For the following Multi-level FSK, find the bitstream from the signal below:

Bit Pattern	Frequency
00	1
01	3
10	2
11	4

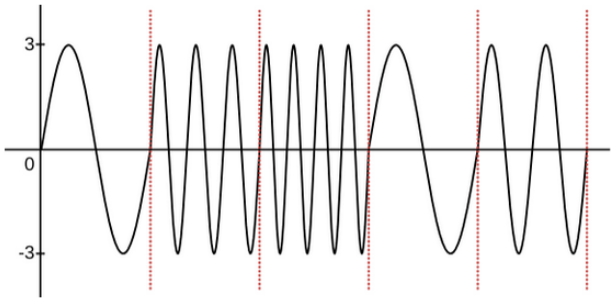


Figure 4: Modulated signal for Question 8

Phase Shift Keying (PSK)

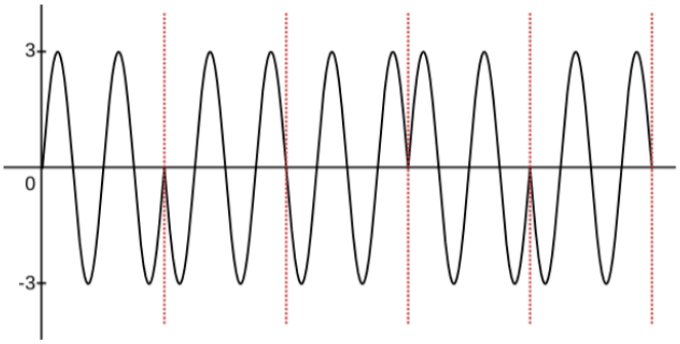


Figure 5: Modulated signal for Question 9

9. Determine the digital bit stream from the analog signal given in Fig. 5. The signal was modulated using Binary PSK, where 0 means signal element with phase of 0 rad and 1 means signal element with phase of π rad.

10. Draw the analog signal for the digital bit stream **X** using Binary PSK, where 0 means signal element with phase of 0 rad and 1 means signal element with phase of π rad. [For each signal element, amplitude = 3V, frequency = 2]

QPSK & Constellation Diagram

11. Draw the constellation diagram for the QPSK given in Fig. 6.

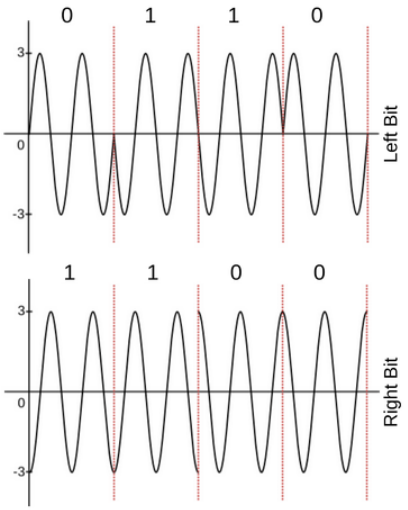


Figure 6: QPSK-modulated signal for Question 11

12. Draw the analog signal for the bit stream **1011001011** using the constellation diagram given in Fig. 7 [Frequency = 2 for each signal element].

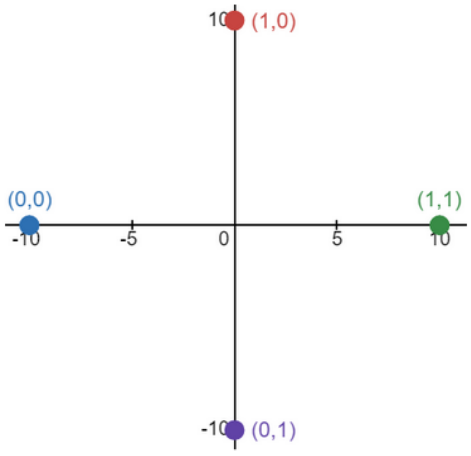


Figure 7: Constellation diagram for Question 12

Frequency Division Multiplexing (FDM)

13. Assume five 20KHz channels are multiplexed using FDM. If the bandwidth of each guard band is 5KHz, calculate the output channel Bandwidth.

14. Assume we have an output channel bandwidth of 100 KHz starting from 1250KHz. There are 5 input channels. If the guard band between two channels is 5KHz, what will be the maximum bandwidth for each channel that we can assign? Draw the frequency domain representation of the required bandwidth. What will be the maximum frequency?

15. Assume we have an output channel bandwidth of 100 KHz starting from 1250KHz. Assume five 20KHz channels are multiplexed using FDM. Calculate the bandwidth of the guard band.

16. The Following FDM hierarchy has been used by a telephone company. How many voice channels can be multiplexed together in the master group? What is the required bandwidth for the multiplexing?

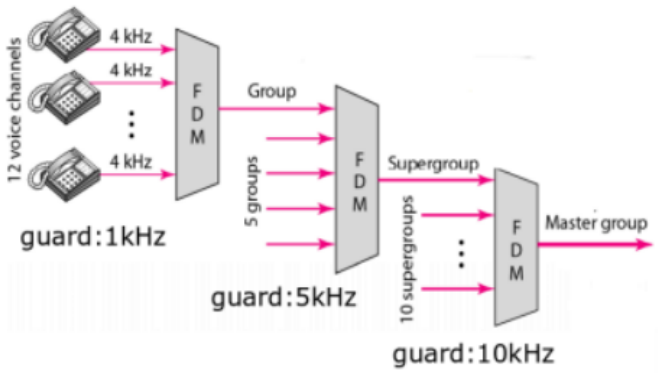


Figure 8: FDM hierarchy for Question 16